

# What a BNF Grammar Defines

A BNF grammar is a *recipe* for building strings.

- Start at the **start symbol** (the first **non-terminal** rule)
- Repeatedly replace a **non-terminal** using a rule (a **production**)
- Stop when you have only **terminals**

The **language** of the grammar is the set of *all* strings you can build this way.

# BNF Grammars

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# Motivating Example

## Writing a JSON Parser

```
{  
  "1": {  
    "location": "assign-ostep7",  
    "weight": 20  
  },  
  "2": {  
    "location": "assign-ostep8",  
    "weight": 20  
  },  
  "3": {  
    "location": "assign-msh3",  
    "weight": 60  
  },  
}
```

(example input text)

Someone asks you to build a JSON parser. How do you approach it?

---

*Input.* A blob of text

*Output.* A dictionary **or** error

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We need a better way to define languages

# Backus-Naur Form (BNF) Grammars

$$\begin{array}{l} R_1 ::= a \mid aR_2 \\ R_2 ::= b \mid bR_1 \end{array}$$

Three parts of grammars:

1. **non-terminal** *symbols* are placeholders that *must* be replaced
2. **terminals** are concrete parts of the output string
3. **productions** are what we replace **non-terminals** with
  - e.g. “**bR<sub>1</sub>**” in the above example



# Using BNFs

To use a BNF:

1. Start with a the start symbol
  - This is the first listed **non-terminal**
2. Replace any **non-terminal** with one of its **productions**
3. Repeat #2 until we have no **non-terminal** symbols left

|       |       |     |     |        |
|-------|-------|-----|-----|--------|
| $R_1$ | $::=$ | $a$ | $ $ | $aR_2$ |
| $R_2$ | $::=$ | $b$ | $ $ | $bR_1$ |

# BNF Practice

# Let's try generating some strings

```
R1 ::= foo | bar
```

# Let's try generating some strings

$R_1 ::= [R_2]$

$R_2 ::= a \mid b \mid c$

# Let's try generating some strings

$R_1 ::= aR_1b$

# Let's try *parsing* some strings

$R_1 ::= aR_1b$

aabb

# Let's try *parsing* some strings

$R_1 ::= aR_1b$

abb

# Let's try *parsing* some strings

$R_1 ::= R_1 + R_1$

abb



# Let's try *parsing* some strings

$$R_1 ::= aR_1b \mid 1 \mid x$$

$1 + x + x$

# Let's try *parsing* some strings

$R_1 ::= aR_1b \mid 1 \mid x$

$1 + x +$